

1-Task Title: Set Operations (A union B, A intersection B, A-B, B-A, A Symmetric Difference B)

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define newl cout<<endl
unordered_map<int,int>C,D;
void Union()
{
    cout<<"A union B->";
    for(int i=0;i<1000;i++)
    {
        if(C[i]||D[i])
        {
            cout<<i<<" ";
        }
    }
    newl;
}
void Intersection()
{
    cout<<"A intersection B->";
    for(int i=0;i<1000;i++)
    {
        if(C[i]==1&&D[i]==1)
        {
            cout<<i<<" ";
        }
    }
    newl;
}
void Exclusive()
{
    cout<<"A symmetric difference B->";
    for(int i=0;i<1000;i++)
    {
        if(C[i]+D[i]==1)
        {
            cout<<i<<" ";
        }
    }
    newl;
}
void Difference(unordered_map<int,int>&A,unordered_map<int,int>&B)
{
    for(int i=0;i<1000;i++)
    {
        if(A[i]==1)
        {
            if(B[i]!=1)
            {
                cout<<i<<" ";
            }
        }
    }
    newl;
}
```

```

}
int main()
{
    cout<<"Total Elements of A Set-> ";
    int a;
    cin>>a;
    cout<<"Total Elements of B Set-> ";
    int b;
    cin>>b;
    vector<int>A,B;
    cout<<"Taking A Set\n";
    for(int i=0;i<a;i++)
    {
        int x;
        cin>>x;
        if(C[x]==0)
        {
            A.push_back(x);
            C[x]++;
        }
    }
    cout<<"Taking B Set\n";
    for(int i=0;i<b;i++)
    {
        int x;
        cin>>x;
        if(D[x]==0)
        {
            B.push_back(x);
            D[x]++;
        }
    }
    Union();
    Intersection();
    Exclusive();
    cout<<"A-B->";
    Difference(C,D);
    cout<<"B-A->";
    Difference(D,C);
}

```

Output:

```

Total Elements of A Set-> 3
Total Elements of B Set-> 3
Taking A Set
1 2 3
Taking B Set
3 4 5
A union B->1 2 3 4 5
A intersection B->3
A symmetric difference B->1 2 4 5
A-B->1 2
B-A->4 5

Press any key to continue . . .

```

Discussion:

As C++ has a STL 'set', to continue set operations I have tried to implement in scratch. Set does not carry duplicate elements so I have used unordered_map to avoid storing duplicate member. A (XOR) B is a new item, I have learnt from this Lab. It is called Symmetric Difference. In this operation, matched elements will be removed.

2-Task Title: Power Set Generate

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define newl cout<<endl
unordered_map<int,int>C;
void power(vector<int>&A)
{
    int n=A.size();
    for(int i=pow(2,n)-1;i>=1;i--)
    {
        int a=i;
        for(int j=0;j<n;j++)
        {
            if(od(a))
            {
                cout<<A[j]<<" ";
            }
            a/=2;
        }
        newl;
    }
    cout<<"empty";
}
int main()
{
    cout<<"Total Elements of A Set-> ";
    int a;
    cin>>a;
    vector<int>A;
    cout<<"Taking A Set\n";
    for(int i=0;i<a;i++)
    {
        int x;
        cin>>x;
        if(C[x]==0)
        {
            A.push_back(x);
            C[x]++;
        }
    }
    newl;
    power(A);
}
```

Output:

```

Total Elements of A Set-> 3
Taking A Set
8 4 9

8 4 9
4 9
8 9
9
8 4
4
8
empty
Press any key to continue . . .

```

Discussion:

To generate power set of a set, I have used bit manipulation techniques to solve. In bit manipulation if the number is odd then the LSB of that number's bit is 1 and pick that indexed member of that set. Total member of a power set will be 2^n where n is the total member of a set.

3-Task Title: Set Multiplication (A*B)

Code:

```

#include<bits/stdc++.h>
using namespace std;
#define newl cout<<endl
unordered_map<int,int>C,D;
int main()
{
    cout<<"Total Elements of A Set-> ";
    int a;
    cin>>a;
    cout<<"Total Elements of B Set-> ";
    int b;
    cin>>b;
    vector<int>A,B;
    cout<<"Taking A Set\n";
    for(int i=0;i<a;i++)
    {
        int x;
        cin>>x;
        if(C[x]==0)
        {
            A.push_back(x);
            C[x]++;
        }
    }
    cout<<"Taking B Set\n";
    for(int i=0;i<b;i++)
    {
        int x;
        cin>>x;
        if(D[x]==0)
        {
            B.push_back(x);

```

```

        D[x]++;
    }
}
cout<<"{ ";
for(int i=0;i<A.size();i++)
{
    for(int j=0;j<B.size();j++)
    {
        cout<<"("<<A[i]<<","<<B[j]<<"),";
    }
}
cout<<"\b }";
}

```

Output:

```

Total Elements of A Set-> 3
Total Elements of B Set-> 3
Taking A Set
1 2 3
Taking B Set
4 5 6
{ (1,4),(1,5),(1,6),(2,4),(2,5),(2,6),(3,4),(3,5),(3,6) }
Press any key to continue . . .

```

Discussion:

In set multiplication, every element of the first set will make a tuple with all the members of the second set. In multiplied set, in every tuple the first element will come from first set and second element will come from second set. If A set has n elements and B set has m elements, then total number of tuple in multiplied set will be $n*m$.